SAP S/4HANA MM — Material Ledger and OBYC

Actual Costing · Multi-Currency Valuation · Automatic Account Determination

Düsseldorf Manufacturing AG (DMAG) — Plants DM01 / DM02 / DM03

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This document focuses on two advanced SAP S/4HANA MM areas; Material Ledger (actual costing, multi-currency valuation) and OBYC automatic account determination, demonstrating my capability in FI-MM integration and period-end processes. It showcases hands-on configuration, costing run execution, variance analysis, and GL account mapping across procurement, inventory, and production scenarios for a multi-plant manufacturing environment.

Part 1 — Material Ledger

1.1 Overview

Material Ledger (ML) is mandatory in SAP S/4HANA. It serves two primary purposes:

- Multi-currency inventory valuation: stock values recorded simultaneously in two parallel currencies (EUR and USD for DMAG)
- Actual costing: calculates the true cost of goods at period end by accumulating all price variances from procurement and production

In S/4HANA, ML captures all variances throughout the period and distributes them automatically at period end via the CKMLCP costing run — giving finance teams actual cost visibility across all plants.

1.2 DMAG Material Ledger Design

Element	DMAG Setting	Rationale
ML Type	ZDM_ML — Actual Costing	Enables true cost accumulation across procurement and production
Plants Activated	DM01, DM02, DM03	All three plants require consistent actual costing for group reporting
Company Code Currency	EUR — Type 10	German statutory reporting currency
Group Currency	USD — Type 30	Parent company consolidation currency
Price Control (ROH/HALB)	V — Moving Average Price	Reflects actual market price fluctuations for raw materials
Price Control (FERT)	S — Standard Price	Enables variance analysis between planned and actual production cost
Costing Run Frequency	Monthly — CKMLCP	Period-end actual cost determination and inventory revaluation

1.3 Activation Steps

Step	T-Code	Activity	DMAG Configuration
1	**OMX2**	Define Valuation Areas	Valuation areas defined for plants DM01, DM02, DM03
2	**OMX3**	Assign ML Type to Valuation Area	ZDM_ML assigned to all three valuation areas
3	**CKMSTART**	Activate and Initialize Material Ledger	ML activated — status PRODUCTIVE. Opening balances set.
4	**OBY6**	Configure Currency Types	Type 10: EUR (Company Code) | Type 30: USD (Group)
5	**OMRW**	Set Valuation Method	Price determination 3 — Single and Multi-Level Actual Costing

6	**CKMLCP**	Execute Actual Costing Run	Period-end run — actual cost calculated and inventory revalued
7	**CKM3N**	Material Price Analysis	Review actual vs standard cost per material, per plant, per period

SAP System Evidence *T-Code: CKMSTART / CKM3N*	
Navigation: *SPRO → Controlling → Product Cost Controlling → Actual Costing/Material Ledger → Activate Valuation Areas*	
ML Status:	`DM01 PRODUCTIVE	DM02 PRODUCTIVE	DM03 PRODUCTIVE`
ML Type:	`ZDM_ML	Price Determination: 3 (Actual Costing)`
Currencies:	`Type 10: EUR	Type 30: USD`
Steel Sheet:	`Standard 2.40 EUR/KG	Actual 2.51 EUR/KG after CKMLCP`
Machined Housing:	`Standard 3.20 EUR/PC	Actual 3.30 EUR/PC after CKMLCP`
What to verify: *All three plants show PRODUCTIVE status. CKM3N shows parallel EUR and USD values with actual cost after costing run.*	

1.4 Actual Costing Scenario — Steel Sheet to Machined Housing

This scenario demonstrates how Material Ledger accumulates actual costs from procurement through production.

Stage	Activity	EUR Cost	USD Parallel	ML Impact
1. Procurement	Steel Sheet purchased at 2.50 EUR/KG — standard price 2.40 EUR/KG	2,500 EUR	2,720 USD	Variance 100 EUR captured by ML
2. Production Issue	Steel Sheet issued to production order for Machined Housing	2,500 EUR actual	2,720 USD	Actual material cost to WIP
3. Processing	Machining activity confirmed on production order: 0.80 EUR/PC	800 EUR	870 USD	Activity cost added to order
4. GR to Stock	Machined Housing received to stock at standard price 3.20 EUR/PC	3,200 EUR std	3,480 USD	Variance 100 EUR held in ML
5. Period End	CKMLCP run — variances rolled up, inventory revalued to actual cost	3.30 EUR/PC actual	3.59 USD/PC	Revaluation posted via OBYC

1.5 Multi-Currency Valuation

Currency Type	Currency	Exchange Rate	Example: 1,000 KG Steel Sheet
10 — Company Code Currency	EUR	Base currency	2,500.00 EUR
30 — Group Currency	USD	1 EUR = 1.088 USD	2,720.00 USD

Exchange rates maintained in OB08. Applied automatically at each posting. Both currencies are visible in all ML reports and FI documents.

Part 2 — OBYC: Automatic Account Determination

2.1 Overview

OBYC (Configure Automatic Postings) maps SAP MM goods movements to Financial Accounting GL accounts. Without OBYC, every goods movement would require manual account selection. The transaction key, valuation class and account grouping determine which GL account is posted automatically.

The valuation class on the material master is the primary driver for BSX (inventory) postings. Different material types post to different inventory GL accounts.

2.2 DMAG GL Account Structure

GL Account	Account Name	OBYC Key	Valuation Class / Trigger
130000	Raw Material Inventory — Steel Sheet, Aluminum Rod	BSX	3000 — Raw Material
131000	Semi-Finished Goods — Machined Housing	BSX	7920 — Semi-Finished
132000	Finished Goods — Pump Assembly	BSX	7900 — Finished Goods
191100	GR/IR Clearing Account	WRX	All material types — both sides of GR and MIRO
281000	Purchase Price Variance	PRD	Standard price materials — invoice vs PO difference
400000	Material Consumption — Production	GBB-VBR	GI to production order (261)
401000	Material Consumption — Cost Centre	GBB-VBO	GI to cost center (201)
410000	Freight Clearing Account	FRE	Freight condition at GR posting
160000	Consignment Payable — Vendor	KON	Withdrawal from consignment stock (201)
170000	Inventory Difference Account	INV	Physical inventory difference posting (MI07)

2.3 Transaction Key Detail

BSX — Inventory Posting

Triggered by every goods movement that changes stock quantity and value: GR (101), GI (261), STO (351/101), return to vendor (122), transfer posting (301). Account selected by valuation class from material master.

Movement	Dr Account	Cr Account
GR (101)	130000 Raw Material Inventory	191100 GR/IR Clearing (WRX)
GI to Production (261)	400000 Consumption-Production (GBB-VBR)	130000 Raw Material Inventory
Return to Vendor (122)	191100 GR/IR Clearing (WRX)	130000 Raw Material Inventory

WRX — GR/IR Clearing Account

The GR/IR account (191100) is a bridge between GR and MIRO. It should net to zero when GR and invoice quantities match. Open items on WRX indicate unmatched GRs or invoices.

Event	WRX Posting	Net Result
At GR (101)	Cr 191100 GR/IR Clearing: 2,450.00 EUR	Open item created — liability to vendor
At MIRO	Dr 191100 GR/IR Clearing: 2,450.00 EUR	Open item cleared — vendor account credited
Net (matched)	191100 balance = 0.00 EUR	Accounting cycle complete for this delivery

PRD — Price Differences

Triggered when invoice price differs from PO price for standard price (S) materials. For Pump Assembly FERT (standard price 8.50 EUR/PC, invoice 8.70 EUR/PC):

Posting Event	Dr	Cr
GR posting	132000 Finished Goods (BSX): 8,500.00	191100 GR/IR Clearing (WRX): 8,500.00
MIRO posting	191100 GR/IR: 8,500 | PRD 281000: 200	Vendor Account: 8,700.00

GBB — Offsetting Entries for Goods Issues

GBB account selection uses both the account modification key (VBR, VBO etc.) and the valuation class. Allows different consumption accounts per material type and purpose.

Sub-Key	Movement	Dr Account	Cr Account
GBB-VBR	GI to Production (261)	400000 Consumption-Production	130000 Inventory (BSX)
GBB-VBO	GI to Cost Centre (201)	401000 Consumption-Cost Centre	130000 Inventory (BSX)
GBB-BSA	Initial Stock Entry (561)	130000 Inventory (BSX)	Offsetting account

KON — Consignment Payable

Unlike standard stock, consignment GR creates no FI posting — stock is recorded but vendor retains ownership. Liability only arises at consumption.

- Consignment GR (101): No FI document — stock recorded in consignment stock type
- Consignment GI (201): Dr 400000 Consumption | Cr 160000 Consignment Payable (KON)
- Periodic Settlement (MRKO): 160000 Consignment Payable cleared to vendor account for payment

SAP System Evidence *T-Code: OBYC*	
Navigation: *SPRO → Materials Management → Valuation and Account Assignment → Account Determination Without Wizard → Configure Automatic Postings*	
BSX Val 3000:	`→ GL 130000 Raw Material Inventory`
BSX Val 7920:	`→ GL 131000 Semi-Finished Inventory`
BSX Val 7900:	`→ GL 132000 Finished Goods Inventory`
WRX (all mats):	`→ GL 191100 GR/IR Clearing Account`
PRD (std price):	`→ GL 281000 Purchase Price Variance`
GBB-VBR:	`→ GL 400000 Consumption Production`
KON (all mats):	`→ GL 160000 Consignment Payable`
INV (all mats):	`→ GL 170000 Inventory Difference`
What to verify: *Transaction key BSX showing valuation class to GL mapping. WRX single account 191100 for all materials. PRD to 281000. GBB sub-keys with separate consumption accounts.*	

Part 3 — Period-End Closing Sequence

Step	T-Code	Activity	ML / OBYC Impact
1	**MB5S**	GR/IR Clearing Review	Identify all open GR/IR items — match uncleared GRs with missing invoices
2	**MMPV**	MM Period Close	Close posting period — prevent backdated postings to previous period
3	**KO88**	Production Order Settlement	Settle production order variances to cost of goods manufactured — feeds ML
4	**CKMLCP (test)**	Preliminary Costing Run	Test run — preview actual cost calculations without posting
5	**CKMLCP (final)**	Actual Costing Run	ML calculates actual cost. Posts revaluation entries via OBYC BSX and PRD.
6	**CKM3N**	Material Price Analysis	Verify actual cost per material, per plant. Review variance layers and cost component split.

SAP System Evidence *T-Code: CKMLCP*	
Navigation: *SAP Easy Access → Accounting → Controlling → Product Cost Controlling → Actual Costing → Edit Costing Run*	
Costing Run:	`DM_OCT2025 | Period: 10/2025 | Company Code: DM01`

Steps Completed:	`Selection → Sequence → Single-Level → Multi-Level → Revaluation → Post Closing`
Steel Sheet:	`Actual 2.51 EUR/KG vs Standard 2.40 | Variance 110 EUR revalued`
Machined Housing:	`Actual 3.30 EUR/PC vs Standard 3.20 | Variance 100 EUR revalued`
FI Impact:	`Revaluation posted to BSX and PRD accounts — inventory restated to actual cost`

What to verify: *CKMLCP log shows all six steps completed without errors. CKM3N confirms actual price update for all materials in all plants.*